37. Given : de Broglie wavelength of the tennis ball equal to 5400 Å.

m = 54 g

v = ?

$$\lambda = \frac{h}{mV}$$

$$V = \frac{h}{m\lambda}$$

$$V = \frac{6.626 \times 10^{-34}\,JS}{54 \times 10^{-3}\,Kg \times 5400 \times 10^{-10}\,m}$$

$$V = 2.27 \times 10^{-26}\,ms^{-1}$$

38.

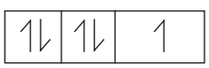
n	l	Sub Enery levels	m_l values	Number of orbitals
4	2	4d	–2, –1, 0, +1, +2	five 4d orbitals
5	3	5f	–3, –2, –1, 0, +1, +2, +3	seven 5f orbitals
7	0	7s	0	one 7s orbitals

41. no. of electrons : 35 (given)

no. of protons : 35

Electronic configuration

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$



$4P_x$ $4P_y$ $4P_z$

last electron present in $4P_y$ orbital

n = 4 l = 1 m_l = either +1 or -1 and s = –1/2

43. $He^+ \rightarrow He^{2+} + e^-$

$$E_n = \frac{-13.6z^2}{n^2}$$

$$E_1 = \frac{-13.6\,(2)^2}{(1)^2} = -54.4$$

$$E_\infty = \frac{-13.6(2)^2}{(\infty)^2} = 0$$

∴ Required Energy for the given process = $E_\infty - E_1 = 0 - (-54.4) = 54.4$ ev

44.

	Atom	Uni-negative ion
number of electron	x – 1	x
number of protons	x – 1	x –1
number of neutrons	y	y

Given that, y = x + 11.1% of x

$$= \left(x + \frac{11.1}{100}x\right) = x + 0.111x$$

$$y = 1.111x$$

mass number = 37

number of protons + number of neutrons = 37

(x – 1) + 1.111x = 37

x + 1.111x = 38

2.111x = 38

$$x = \frac{38}{2.11}$$

x = 18.009 x = 18 (whole number)

∴ Atomic number = x – 1

= 18 –1 = 17

Mass number = 37

Symbol of the ion $^{37}_{17}Cl^-$

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